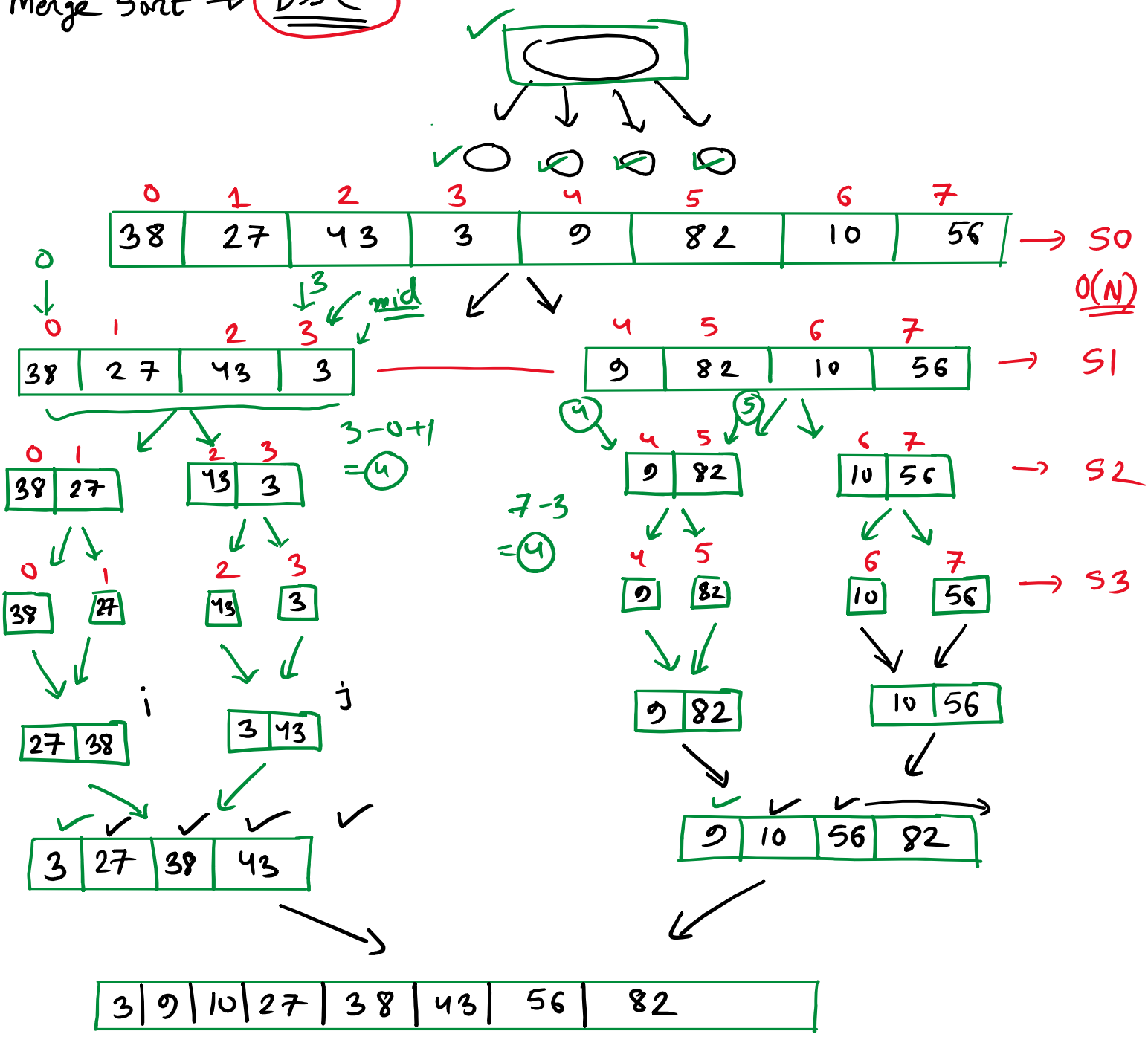
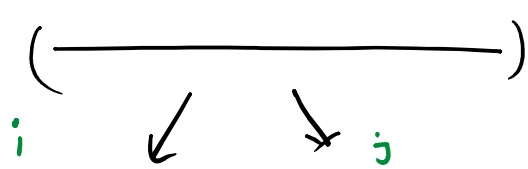
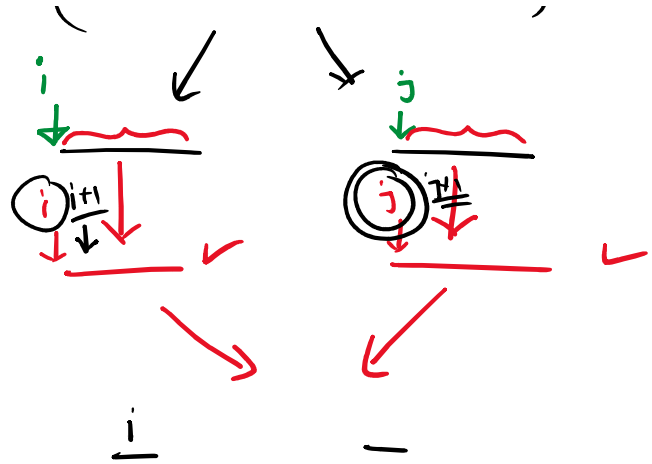
Merge Sort and Quick Sort

Thursday, April 23, 2026 9:00 PM

Merge Sort → DSC







$[N, N/2, N/4, N/8, \ldots, 4, 2, 1]$

$[1, 2, 4, \ldots, N/2, N]$

$[2^0, 2^1, 2^2, \ldots, 2^{S-1}, 2^S]$

$$\boxed{2^{step} = N}$$

$$\boxed{step = \log_2(N)}$$

$$N \times \frac{(\text{number of steps})}{\log_2 N}$$

$O(N \log N)$

$[8, 4, 2, 1]$

$[1, 2, 4, 8]$

$[2^0, 2^1, 2^2, 2^3]$

$$2^{step+1} = N$$

$$step = \log(N) - 1$$

$O(N \log N)$

pivot ⇒

arr[j] ≤ 42 (Pivot)

				i			j
23	9	18	32	42	50	78	61
0	1 ↑	2	3	$\boxed{4}$	5	6 ↑	7

$\boxed{1}$, 5, 10, 12, 15, 20

n → n-1 → n-2 → ..... → 1

5 → 4 → 3 → 2 → 1

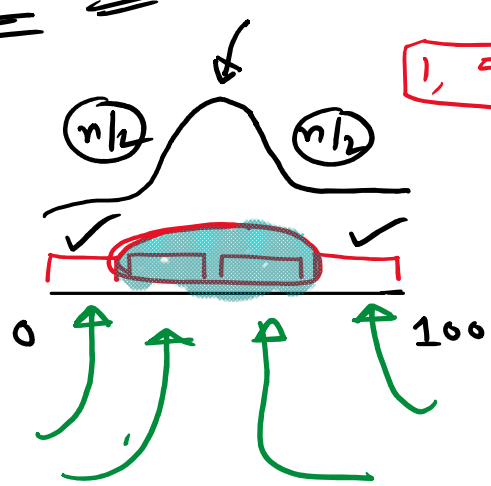
$\boxed{20}$

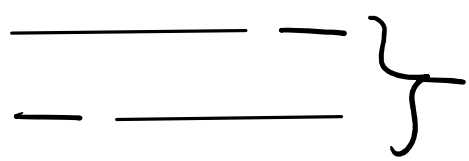
$\boxed{1, 5, 10, 12, 15}$

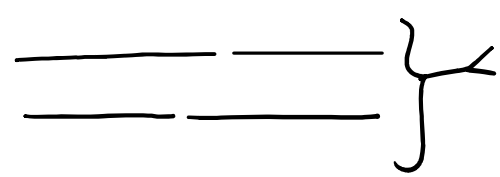
15

$\boxed{1, 5, 10, 12}$

$O(N^2)$





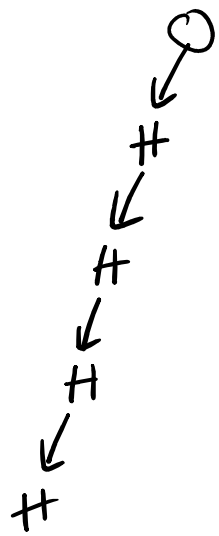


$O(N \log N)$

$O(N^2)$

N, N-1, N-2 ..... $O(N^2)$

$O(N^2)$ $\qquad$ $N, N-1, N-2 \ldots\ldots$ $O(N^2)$



x

$(1, 3)$

$(1, 7)$

$(1, 0)$

$(2, 1)$

$(0, -1)$

$(0, 10)$

merge sort

(stable)

$(0, -1)$

$(0, 10)$

$(1, 3)$

$(1, 7)$

$(1, 0)$

$(2, 1)$